COAL QUIZ 3 (A)

NAME: ____________________________ SECTION______ STUDENT ID: _____________

1. Update flags after arithmetic instruction. Also state which of the following jumps will taken or not taken

```
.code

   mov ax,0FFFEh
   add ax,0FC70h
    jc l1
L1: jz L2
L2: jo L3
L3: js L4
L4: jp L5
L5:
```

	TAKEN	NOT TAKEN
Jc		
Jz		
Jo		
Js		
jp		

FLAGS	
Sign	
Zero	
Carry	
Overflow	
Parity	

Calculations

2. Update flags after every **CMP** instruction and calculate jumps

```
MOV ax,12
CMP cx,-12
JA L1
JNBE L2
JNLE L3
JG L4
```

	TAKEN	NOT TAKEN
JA		
JNBE		
JNLE		
JG		

Subtraction to Calculate Flags for unsigned numbers

2's Complement addition to calculate flags for signed number

3. Update register after every instruction

```
mov ax,0AD14H
mov bl,25
xor al,-1
xor bl,al
xor al,bl
xor ah,ah
```

al	
bl	
al	
ah	

COAL QUIZ 3 (A)

NAME: ____________________________ SECTION______ STUDENT ID: ____________

4. Convert the following **if condition** to equivalent Assembly code using conditional jumps

C Code	Assembly

`int al=125;`
` if(al<=-5 || al>=127)`
` {`
`     al=255;`
` }`
` else`
` {`
`     al=-128;`
` }`
`}`	

5. Convert the following Assembly code equivalent **C Code.**

Assembly	C Code

`mov eax,0`
`mov ebx,0`
`mov ax,-5`
`l1:`
`   inc bl`
`   add al,3`
`   cmp al,5`
`jl l1`	

6. Dry run given C code and write final values of all integers.Convert the following **loops** to equivalent Assembly code using conditional jumps

C Code	Assembly

`for(int al=-5;al<11;al=al+5)`
` {`
`    int bl=40,cl=0;`
`    while(bl>=27)`
`     {`
`       bl=bl-4;`
`       cl++;`
`     }`
` }`	

Al	
Bl	
Cl	